Multifractal Spectrum Analysis of Metal-Organic Framework Networks via Topology-Based Coarse-Graining

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REDLINE VERSION. red highlight = text removed green highlight = text added Unmarked text is the original. A clean version with all changes accepted is provided separately.

Metal-organic frameworks (MOFs) are porous crystals whose behavior depends strongly on how their building blocks are connected. Simulating gas adsorption for every candidate framework is expensive, so a cheap structural descriptor that can rank frameworks before any simulation would be useful. In this work we build a pipeline that turns a MOF crystal structure into a coarse-grained graph of metal-oxo nodes and organic linkers, and then measures the heterogeneity of that graph with an influential-node multifractal analysis. The width of the resulting multifractal spectrum, $\Delta\alpha$, is used as the descriptor. We ran the pipeline on 150 structures returned by a live MOFX-DB query. The query asked for CoRE MOF 2019 entries with xenon isotherms, but the database returned hypothetical hMOF structures, which our provenance check flagged. Of these, 76 gave a usable spectrum. The values of $\Delta\alpha$ ranged from 1.032 to 1.582, with a narrow interquartile band and a tail of wide-spectrum frameworks. We also report three problems that limit the current results: $\Delta\alpha$ depends strongly on graph size, several structures are topological duplicates, and individual power-law fits are often poor. We have not yet tested whether $\Delta\alpha$ predicts any adsorption property. A companion analysis shows that the size dependence is exact: the width grows with the logarithm of the number of graph vertices, so it cannot be a material property. Connectivity alone is also not sufficient to describe adsorption, because it cannot see chemical identity or the geometry of the building blocks. We therefore add chemical descriptors to the plan and extend the work to a mathematical and computational study of framework morphism (for example tetrahedral, spherical and rod-like building-block shapes), the direction requested by our review panel.

Index Terms—Metal-organic frameworks, multifractal analysis, coarse-graining, network topology, MOFX-DB, hMOF, chemical descriptors, framework morphism.

I. INTRODUCTION

Metal-organic frameworks (MOFs) are crystalline materials made from metal-containing nodes joined by organic linkers. Changing the node, the linker, or the way they connect gives a very large family of structures, and many of them are studied for gas storage and separation. The number of possible frameworks is far bigger than what anyone can simulate. Grand canonical Monte Carlo (GCMC) runs for adsorption are accurate but slow, and running them on every structure in a large database is not realistic.

A common way to deal with this is to filter first. If some inexpensive number computed from the structure alone could sort frameworks into groups, the expensive simulations could be limited to a smaller subset. Most simple descriptors of this kind, such as void fraction or pore diameters, come from the geometry of the empty space. In this project we look at a different side of the problem, which is the connectivity of the solid framework itself. We ask how uneven or heterogeneous the network of building blocks is, and whether a single multifractal number can capture that.

The idea is to represent each MOF as a graph. A single unit cell can have hundreds of atoms, and an atom-level graph mostly reflects bond chemistry inside each linker. Grouping atoms into chemically meaningful blocks, which is the coarse-graining step, gives a much smaller graph where each node is either a metal-oxo cluster or a whole organic linker. We then apply influential-node multifractal analysis (iNMFA) to this coarse-grained graph and use the width of the spectrum, $\Delta\alpha$, as a descriptor.

Connectivity alone, however, is only part of what controls adsorption. Which metal and which linker chemistry are present, how polar and how charged the pore surface is, and how large and what shape the pores are all matter, and none of them is a property of the bare graph. We therefore treat the topological descriptor as one layer of a longer pipeline, and Section VI-A explains why chemistry must be added to it.

This paper describes what we built and what we found. We want to be clear from the start about two things. First, our data query did not return what we asked for, so all results here come from hypothetical hMOF structures and not experimental CoRE MOF 2019 structures. Second, the descriptor is not yet clean. We found that $\Delta\alpha$ changes a lot with the size of the graph, and we describe this in detail in Section V. Because of this, we treat the current work as a working pipeline with an honest list of open problems, and not as a finished screening tool.

The contributions are as follows.

1. A reproducible pipeline from a live MOFX-DB query to a coarse-grained building-block graph, using a simple metal-oxo rule.

2. An application of iNMFA to the coarse-grained supercell graph, with repeated randomized box coverings to reduce noise.

3. A data-driven grouping of 76 frameworks into Narrow, Medium and Wide classes by $\Delta\alpha$.

4. A list of the diagnostics that exposed the weaknesses of the method, including the finite-size effect, duplicate structures, and unstable fits.

5. An argument, supported by our own calculations, that topology alone is not sufficient, and a plan that adds chemical descriptors and studies framework morphism (Sections VI-A, VI-B and VII).

II. BACKGROUND AND RELATED WORK

A. Databases

CoRE MOF 2019 is a curated collection of experimental MOF crystal structures that were cleaned for computational use [3]. MOFX-DB collects structures together with simulated adsorption data and offers a web interface and a Python client [4]. The hMOF database is a large set of hypothetical MOFs generated computationally by combining building blocks [5]. Hypothetical structures are useful for screening studies, but they are not experimentally synthesized, and they are built from a limited library of blocks. This matters for our results, as discussed later.

B. Building-block extraction

MOFid gives a standard way to identify the metal nodes and organic linkers of a MOF [2]. MOFDiff uses coarse-grained building blocks as the representation for a generative diffusion model for MOF design [1]. In MOFDiff, each block is described by a set of connection points, and block identity is learned by a neural network. Our work borrows the general idea of splitting a framework into nodes and linkers, but we do not implement either MOFid or MOFDiff. Our partition rule is simpler and it is described in Section IV.

C. Fractal and multifractal analysis of networks

Box covering is a standard way to study self-similarity in networks. The network is covered with the smallest practical number of boxes of a given size, and the way the box count changes with box size is examined [7]. Multifractal analysis goes further by assigning a probability mass to each box and studying moments of that distribution at different exponents q [8]. If the mass distribution is uneven across the network, the resulting spectrum is wider. ; however, for a finite graph the width also grows with the number of vertices (Section V-D), so a wide spectrum is not by itself evidence of an uneven structure. Influential-node multifractal analysis applies this idea only to a chosen subset of high-importance nodes, which reduces the cost and focuses the analysis on the hubs of the network. Our implementation follows the standard formalism, with the specific choices listed in Section IV.

Table 1: Number of structures at each stage.

Stage	Count
Records returned by MOFX-DB	150
Connected periodic atom graph	77
Successful iNMFA spectrum	76
Failed (radius range too small)	1

D. Topology and chemistry in recent MOF models

Chen et al. [10] split a structure by element and compute persistent topology for each element subset and for pairs of elements, and then pass these features to a transformer. They report that adsorption and transport depend mostly on pore topology, while the band gap relies more on which atoms are present, so topological features are used together with element-aware ones. Anselmi et al. [11] train a graph transformer on bond, line and Coulomb graphs to screen MOF photocatalysts and find that the prediction error depends on the metal: for lanthanide-containing structures the error is lowest on elements seen in training (0.18 eV) and highest on unseen ones (1.09 eV). Both works carry chemical identity in the representation. Our block graph does not.

III. DATA

All structures were requested live from MOFX-DB through the mofdb-client Python package. The query asked for the database named CoRE MOF 2019 and required a xenon isotherm. We used the incremental get_all generator and stopped after 150 records, giving a candidate pool sorted by record identifier. No individual structure was selected by hand.

After download, our code checks the database field of each returned record against the requested one. This check showed that the server returned hMOF records and not CoRE MOF 2019 records. We do not know the reason for this mismatch. It may be a problem with how our query parameters are passed, a change in the server, or an issue in the client library. We have not yet resolved it. Because of this, all frameworks analyzed in this paper should be read as hypothetical hMOF structures.

Each record carries its CIF text. We write it to disk and read it with ASE [6]. A record is rejected if the CIF is empty, if the cell is not fully periodic, or if it has fewer than four atoms. Table I shows how the set shrank across the pipeline.

The drop from 150 to 77 is large. Those structures were removed at the parsing, connectivity, or supercell stages, and we did not keep a clean per-stage breakdown in the summary we are reporting from. The connectivity check is strict, and a disconnected periodic graph often points to an interpenetrated framework or leftover guest molecules in the CIF. We think this is the main reason, but we have not verified it structure by structure.

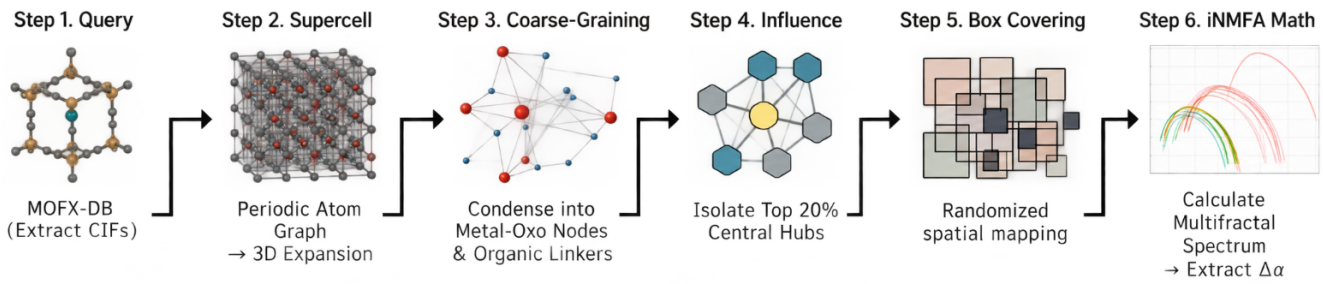


Figure 1: Overview of the computational pipeline. Starting from a MOF structure retrieved from MOFX-DB, a periodic atomistic bond graph is constructed and expanded into a supercell. The structure is then coarse-grained into a building-block graph of metal-oxo nodes and organic linkers. The top 20 influential blocks are selected, The top 20 percent of blocks by degree are selected as influential nodes, and influential-node multifractal analysis (iNMFA) is performed via randomized box covering to obtain the multifractal spectrum and its width $\Delta\alpha$.

IV. METHOD

A. Periodic atom graph

Bonds are found with the ASE neighbor_list function under full periodic boundary conditions, using a cutoff that depends on the element pair. The cutoffs are 2.80 Å for metal-metal pairs, 2.45 Å for metal-nonmetal pairs, 1.70 Å for pairs made of C, N and O, 1.45 Å for any pair containing H or F, and 1.90 Å for everything else. The metal set is a fixed list of 33 elements common in MOF nodes. These cutoffs are practical choices and not fitted values. A structure whose periodic unit-cell graph is not connected is dropped.

B. Supercell

Each unit cell is repeated until every lattice vector is at least 48 Å long, with a cap of 7000 atoms so that the all-pairs shortest-path computation stays affordable. If the cap would be exceeded, the repeat count is reduced one direction at a time, starting from the direction that contributes the most atoms. The atom graph is rebuilt on the supercell with the same cutoffs and must again be connected. The supercell is built before coarse-graining. It gives the multifractal analysis a larger range of graph distances to work with.

C. Coarse-graining

The partition is done on the supercell atom graph. Every metal atom is put in a node set together with every oxygen atom directly bonded to it. The connected components of the subgraph induced by this set are the metal nodes. All remaining atoms are treated as organic, and the connected components of their induced subgraph are the linkers. Every atom therefore belongs to exactly one block. The coarse-grained graph G_{cg} has one vertex per block, and two blocks are joined if at least one atom-level bond crosses between them. No coordinates are computed for the blocks.

This rule is much simpler than MOFid. It only adds oxygen atoms one bond away from a metal, so more complex clusters can be split or mis-assigned. It also does no bond cutting and no embedding, which MOFDiff does. We therefore describe the step as a simplified, MOFid-inspired partition and not as an implementation of either method.

Two remarks from the companion analysis [9]. First, there the metal-oxo rule of MOFid [2] is used: every bond incident to a metal is cut, and fragments that are a single atom or contain only O and H become nodes. The two rules differ for carboxylate oxygens, which join the node here and stay in the linker there, so the two graphs may differ; we have not compared them. Second, for that rule we proved that coarse-graining the supercell is the same as coarse-graining the unit cell and then replicating it, provided all blocks are finite, and we confirmed this on 5 test cases. We expect the same for the rule used here, so that building the supercell first is harmless, but we have not checked it.

A structure is kept only if G_{cg} has at least 12 nodes and 8 edges and is connected.

D. Influential nodes

On the coarse-grained graph with N nodes we compute the all-pairs shortest path distance matrix D (unweighted). Nodes are ranked by degree, with closeness centrality breaking ties. The top 20 percent are taken as influential nodes. We used 20 percent and not the 10 percent we had used on atom graphs, because the coarse-grained graphs are much smaller.

E. Randomized box covering

For a box radius r, the graph is covered by boxes in the following way. Nodes are visited in a random order. If a visited node is not yet covered, it becomes a box center, and all still-uncovered nodes within distance r of it are put into that box. This continues until all nodes are covered. Because the visiting order is random, the covering changes from run to run.

The radii used are $r = 1, \ldots, r_{\max}$, where $r_{\max}$ is half of the graph diameter (rounded), with a minimum of 3 and a maximum of the diameter minus one. Structures with fewer than 3 usable radii are rejected, and this is what removed one framework in our run.

F. Multifractal quantities

For each box i that contains an influential node, we define the probability mass $p_i(r) = n_i(r)/N$, where $n_i(r)$ is the number

of nodes in the box. The partition function is

$$P_q(r) = \sum_i [p_i(r)]^q, \tag{1}$$

where the sum runs over boxes containing influential nodes. We use 41 values of q evenly spaced from -10 to 10. For $q = 0$ the sum is defined as the number of such boxes, to avoid ambiguity with 0^0.

The mass exponent $\tau(q)$ is obtained from

$$\ln P_q(r) = \tau(q) \ln\left(\frac{r}{d}\right), \tag{2}$$

where d is the graph diameter. We fit this as a straight line through the origin, so $\tau(q) = \sum xy / \sum x^2$ with $x = \ln(r/d)$ and $y = \ln P_q$. We use the through-origin form because in earlier tests a free intercept destroyed the peak of the spectrum at $q = 0$. The price is that the fit is forced, and we return to this in the discussion.

Remark. Eq. (1) sums over boxes. The companion analysis [9] sums over the influential vertices, each carrying the mass of its box; the two agree when each box holds at most one influential node. With the vertex sum, $\ln P_0 = \ln |I|$ for every r ($|I|$ is the number of influential nodes), the through-origin fit then fixes the height of the peak of $f(\alpha)$ at $\rho \ln |I|$ with $\rho = \sum |x| / \sum x^2$, whatever the structure, and a free intercept fixes it at zero. Here the peak height depends on the box counts instead, and we have not repeated the comparison.

To reduce randomness, we run 8 independent batches, each averaging $\ln P_q(r)$ over 8 random coverings. Then $\tau(q)$ is fitted once per batch. The singularity strength and the spectrum follow from a numerical derivative and a Legendre transform,

$$\alpha(q) = \frac{d\tau}{dq}, \qquad f(\alpha) = q\,\alpha(q) - \tau(q). \tag{3}$$

The spectrum width is $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$, computed for each batch and then averaged. We also record α_0, taken at the maximum of $f(\alpha)$, and an asymmetry

$$A = \ln \frac{\alpha_0 - \alpha_{\min}}{\alpha_{\max} - \alpha_0}, \tag{4}$$

which is negative when the spectrum is longer on the large-α side.

Gaussian smoothing is applied only to the plots. All reported numbers come from the raw spectra.

G. Classification

Frameworks are sorted into three classes using the 25th and 75th percentiles of the $\Delta\alpha$ values in the set. Values at or below the lower percentile are Narrow, values above the upper percentile are Wide, and the rest are Medium. The thresholds come from the data, so they will move if the dataset changes.

H. Diagnostics

Besides the main run, the code performs a seed check, a finite-size check on one representative framework, and an isomorphism check that compares unit-cell atom graphs of equal size. We also looked at simple statistics of the coarse-grained graphs, namely clustering and degree assortativity.

Table 2: Classes defined by percentiles of $\Delta\alpha$.

Class	Range of $\Delta\alpha$	Count
Narrow	≤ 1.190	19
Medium	1.190 to 1.222	38
Wide	> 1.222	19

V. RESULTS

A. Spectra

All 76 spectra have their maximum of $f(\alpha)$ at $q = 0$, which is the expected shape for a well-behaved multifractal spectrum. This is guaranteed by construction and is not evidence of good scaling: with a through-origin fit, $\tau(q)$ is concave for any positive data (the logarithm of a sum of powers is convex in q and the fit weights x are negative), so $f(\alpha)$ has its maximum at $q = 0$ for every graph. The asymmetry A was negative for every framework. Fig. 2 shows the individual spectra on the left and the class averages on the right.

In the left panel, most Narrow and Medium spectra sit close together with peaks near α of 1.8 and $f(\alpha)$ between 2.1 and 2.4. The Wide spectra are shifted to larger α, with their maxima higher and their right ends reaching about 3.4. A few curves extend much further, and one reaches α close to 4. In the right panel the Narrow and Medium averages overlap over most of the range and cannot be separated cleanly. The Wide average is clearly different, with a much larger spread. The bump near $\alpha = 2.75$ in the Wide average is probably caused by a small number of curves reaching that range, and we do not read anything into it.

B. Distribution of $\Delta\alpha$

Across the 76 frameworks, $\Delta\alpha$ ranges from 1.032 to 1.582. The middle of the distribution is tight, with the interquartile band running only from 1.190 to 1.222. Most frameworks are therefore very similar by this measure, and the Wide class is a tail above that core. Table II lists the classes.

The Narrow and Medium classes are separated by a very small difference in $\Delta\alpha$, since the lower edge of the middle band is 1.190 and the upper edge is 1.222. Percentile classes always give a 1:2:1 split by construction, so the class sizes carry no information. We would not claim that the boundary between Narrow and Medium is meaningful, and the overlap in Fig. 2 supports that caution.

C. Graph statistics

The coarse-grained graphs have clustering close to zero, meaning there are almost no triangles. This is expected for framework networks where blocks link to a small number of neighbors in an open lattice. Degree assortativity is strongly negative, so high-degree blocks tend to connect to low-degree ones. This fits the usual picture of metal nodes with many connections joined by linkers with two. It also means that much of the heterogeneity the method detects may come from the alternating node and linker structure that all frameworks share, and not from the differences between them.

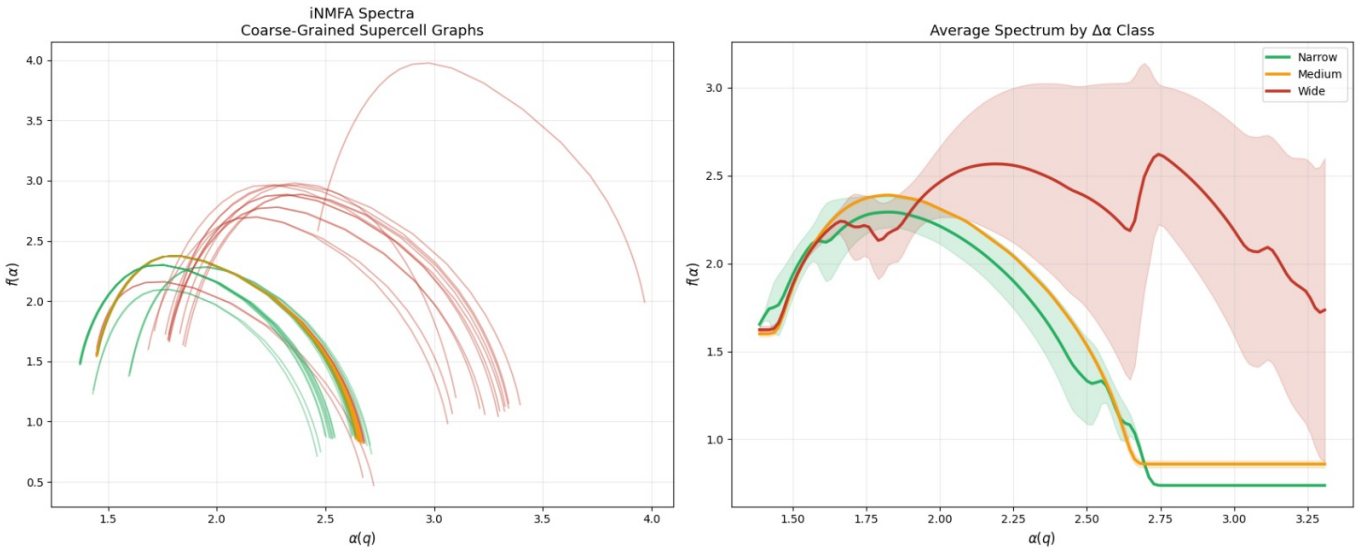


Figure 2: iNMFA results on coarse-grained supercell graphs. Left: individual $f(\alpha)$ spectra, colored by class (green Narrow, orange Medium, red Wide). Right: class-average spectrum with one standard deviation band. Smoothing is for display only. The flat sections at the right end of the Narrow and Medium averages come from filling points outside each curve's range during averaging, so they should not be read as data.

D. Finite-size effect

This was the most important negative result. We repeated the analysis on the representative framework chosen by the code, using different supercell sizes. On the $1\times1\times1$ cell we obtained $\Delta\alpha\approx0.54$, and on the $2\times2\times2$ cell the value was about 1.18. The framework is chemically the same in both cases, since the supercell only tiles the same blocks. So the value of $\Delta\alpha$ is not purely a property of the topology. It also depends on how many nodes are in the graph and on the range of radii that this allows. Any comparison of $\Delta\alpha$ between frameworks whose coarse-grained graphs have different sizes is therefore partly a comparison of graph size. Because our supercell rule fixes edge lengths and not node counts, frameworks with small unit cells and frameworks with large ones end up with different numbers of blocks.

The companion analysis [9] explains this growth exactly (for the vertex sum described in Section IV-F). For $q\to-\infty$ the partition function is dominated by the smallest box mass, and this gives $\alpha(-\infty)=\rho\ln N-\mu$, where $\rho=\sum|x_k|/\sum x_k^2$ is fixed by the set of radii, $x_k=\ln(r_k/d)$ as in Eq. (2), and the term $\mu\geq0$ stayed between 0.54 and 1.04 in our tests. The width therefore grows like $\rho\ln N$. On supercells of HKUST-1, MOF-5 and ZIF-8 (N from 112 to 6912) the value of α at $q=-10$ was within 0.011 of this limit, and $\Delta\alpha$ rose from about 1.2–1.4 to about 2.0–2.3 while the structure stayed the same. A different fit window changes ρ but cannot remove the $\ln N$ term. An independent analysis of 72 hMOF structures also gave a correlation of $r=0.75$ between $\Delta\alpha$ and $\ln N$. We therefore do not read the narrow band of Table II as a property of the structures.

E. Duplicates

The isomorphism check found 5 groups of structures whose unit-cell atom graphs are isomorphic, containing 17 structures in total. These are topological duplicates, at least at the level of the atom graph. The effective number of distinct frameworks is therefore lower than 76. We did not remove them before computing the class thresholds, so the percentiles are also affected. Note that hypothetical databases are built from a limited set of blocks and nets, which makes such repeats more likely. In the independent analysis, 65 of 72 hMOF structures share the same pair of node degree and linker degree (six-connected nodes, two-connected linkers), consistent with many of them being variants of one net.

F. Quality of the fits

The through-origin R^2 for individual q values is often poor. The mean R^2 of a framework was negative in many cases, with one example of -1.17. Negative values are possible because we compute R^2 against the mean of the data while forcing the line through the origin, so a fit can be worse than a horizontal line. It does not necessarily mean that the box-counting data has no scaling. It does mean that individual slices of $\tau(q)$ are noisy, even though the averaged spectra look smooth and well shaped. The smooth look of the spectra should therefore not be taken as proof that each power-law fit is good. The number of radii used is small, especially for the smaller graphs, and this contributes to the problem. In the companion analysis the linearity of $\ln P_q$ against $\ln(r/d)$ is judged by the R^2 of a free-intercept fit, which does not have the negative-value problem, while $\tau(q)$ itself is still taken from the through-origin fit.

VI. DISCUSSION

The pipeline works mechanically. It goes from a database query to a coarse-grained graph to a well-shaped concave (by construction, see Section V-A) multifractal spectrum for 76 structures without manual selection. However, the results do not yet support using $\Delta\alpha$ as a reliable descriptor, for the following reasons.

First, the data are not what we intended. The results describe hypothetical hMOF structures, so we cannot say anything about experimental frameworks. Fixing the query, or switching to another endpoint, is the first thing to do.

Second, the finite-size dependence is severe. A change of about 0.6 in $\Delta\alpha$ between two supercell sizes of one framework is larger than the whole range we observed across the set, which is about 0.55. This means that the differences between frameworks could be dominated by the graph size. A fair comparison needs either graphs of matched size, or a fit window that is normalized in a way that removes this dependence. We have not done either yet. Section V-D shows that a window alone cannot remove it, so matched sizes or a different, size-independent descriptor are needed.

Third, many frameworks look alike. The middle 50 percent of $\Delta\alpha$ values lies in a band of width about 0.03, and 17 structures are duplicates. The descriptor may separate only a small number of unusual frameworks from a large uniform group.

Fourth, we discard geometry after the graph is built. Bond lengths, angles, pore shapes, and the chemistry of the linkers do not enter the graph. Two frameworks with the same connectivity but very different pores would receive the same score. Adsorption depends strongly on pore geometry, so this is a real limitation for the intended use.

Fifth, the coarse-graining rule is simplified. The single-hop metal-plus-oxygen rule can mis-assign atoms in complex clusters, which changes the coarse-grained graph and therefore the spectrum.

Finally, we have not yet checked whether $\Delta\alpha$ correlates with anything measurable. The database provides xenon isotherms, and the void fraction, the largest cavity diameter and the pore limiting diameter are available for the records, so this test is possible. Without it, we cannot say that the descriptor is useful for filtering. The original motivation of the project remains untested.

A. *Why topology alone is not sufficient*

Three observations from our own calculations show that a graph descriptor cannot be enough.

(i) The graph cannot see the shape of a node. Every node block is one vertex. In the companion analysis [9], the Zn_4 core of MOF-5 is a regular tetrahedron (all six Zn–Zn distances 3.17 Å) and the Zr_6 core of UiO-66 is a regular octahedron, yet both appear as a single vertex. Their shape descriptors differ clearly (a third-moment measure is 0.471 for the tetrahedron and about 0 for the octahedron, whereas the usual gyration-tensor anisotropy is zero for both), but the graph treats them alike.

(ii) The graph cannot see chemistry. Two frameworks with the same net but different metals, linkers or functional groups receive the same $\Delta\alpha$, although their pore chemistry, partial charges and interaction with a guest such as xenon, carbon dioxide or water differ. Recent models carry this information explicitly [10], [11].

(iii) Geometry is discarded (the fourth point above).

The chemical layer that must be added consists of pore descriptors (void fraction, largest cavity diameter, pore limiting diameter, accessible surface area, for example with Zeo++ [12]); element identity and atomic partial charges; and adsorption quantities from Widom insertion or GCMC (Henry coefficient K_H, uptake, and selectivity $S_{ij} = K_{H,i}/K_{H,j}$, for example with RASPA [13]). The topological number can then be used as one input among several and tested by correlation with these.

B. *Direction: framework morphism*

Our review panel asked for a mathematical and computational demonstration that a framework can change form, for example from a tetrahedral to a spherical or a rod-like arrangement, since this is known chemically but has not been shown mathematically. The companion analysis [9] gives two parts. Dimensional morphism: removing linkers from the periodic quotient graph of MOF-5 takes it through periodicity rank 3, 2, 1, 0 (a three-dimensional framework, layers, chains, isolated nodes) with 1, 2, 4, 8 connected components, and the number of zero Laplacian eigenvalues of the $2\times2\times2$ supercell is 1, 4, 16, 64. Shape morphism: the gyration-tensor asphericity, a third-moment invariant and Steinhardt bond-orientational invariants separate tetrahedral, octahedral, spherical and rod-like point sets, and an explicit path from a tetrahedron to a rod is computed in closed form. This is a model-level result. It does not show that a real crystal follows such a path, and it contains no energy and no chemical property. The next step is therefore to evaluate the chemistry along the path (Section VII).

VII. FUTURE WORK

The next steps, in the order we plan to do them, are the following.

1. Fix the query so that real CoRE MOF 2019 structures are returned, and confirm this with the provenance check.
2. Remove the size confound, either by comparing graphs with similar numbers of nodes or by changing the fit window, and repeat the finite-size test on several frameworks and not only one. A window alone will not remove it (Section V-D); a size-independent descriptor is preferable.
3. Remove duplicate structures before defining the percentile classes.
4. Test whether $\Delta\alpha$ correlates with pore metrics and with simulated xenon uptake.
5. Improve the fit quality by using more radii, and examine whether a free intercept can be used without losing the peak at $q = 0$.

6. Consider a better partition rule closer to MOFid, and add simple geometric or chemical descriptors alongside the topological one.

7. Compute pore descriptors (Zeo++) and adsorption properties (Widom insertion or GCMC with RASPA) for the same structures, and test them together with element identity, charges and $\Delta\alpha$, treating topology as one input among several.

8. Define framework morphisms (linker deletion; deformation of building-block shape) and follow geometric and adsorption properties along each path, with an energy or constrained optimisation attached to the path.

9. Replace $\Delta\alpha$ by size-independent topological quantities, such as the stiffness tensor of the periodic quotient graph [9], and test whether they carry information beyond pore geometry.

VIII. CONCLUSION

We built a pipeline that takes MOF structures from a live database query, converts them into coarse-grained building-block graphs on supercells, and measures the multifractal spectrum width of the influential nodes. On 76 hypothetical hMOF structures the method gave well-shaped concave spectra that all peak at $q = 0$(by construction), with $\Delta\alpha$ between 1.032 and 1.582 and most values in a narrow band. At the same time, the work showed clear limits. The database did not return the experimental set that we requested, $\Delta\alpha$ changed from about 0.54 to about 1.18 when the supercell of one framework was enlarged, 17 structures were topological duplicates, and individual fits were often poor. For these reasons we regard $\Delta\alpha$ as a descriptor that is not yet validated. The pipeline is a usable base, and the next tasks are to obtain the correct data, remove the dependence on size, and test the descriptor against adsorption data. We also conclude that topology alone is not sufficient: chemical identity and building-block geometry have to be added, and the planned next step is to combine them with a mathematical and computational study of framework morphism.

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